AAVISHKAR

Research Convention

Title of the research project

Integrated Railway Maintenance Prioritization and Constraint-Based Block Planning for Asset Availability

Category	Level	Slot / Code No.
5 — Engineering and Technology	_____	_____

Maintenance is still requested department by department

What is fragmented

Engineering, traction, and signalling keep defect records in TMS, SMMS, and TDMS. Each department asks for its own block through BDMS. COA holds corridor availability. The timetable is a separate document. [1]

Why the gap matters

Mail/Express punctuality on Indian Railways was 73.62% in 2023–24 and 77.12% in 2024–25. Asset maintenance is one stated cause of delay. [2] In a 2017 audit of selected heavy-traffic sections, non-availability of blocks was the stated reason deep screening fell overdue. [4]

Research objective

Can separate maintenance records and a working timetable be turned into one explainable priority, a separate escalation forecast, shared possessions, and a feasible block a person still has to approve?

Rolling Block, notified on 30 November 2023, already asks for integrated maintenance to be planned ahead. This prototype tests one way to do that on a modelled corridor. [3]

METHOD

One pipeline. The numbers are not merged.

TMS, SMMS, TDMS, COA, and WTT are the intended feeds. In this build they are file readers plus an encoded CSMT–Pune model, not live railway servers.

TMS Track	SMMS Signalling	TDMS Traction	COA Corridor	WTT Timetable

1 Read and check

Parse each export. A bad row is rejected with a reason.

2 AI-1 review

Local rules propose findings. A quote must be on the page. A person applies ready rows.

3 Formula 1

Explainable urgency from severity, overdue days, TSR, and traffic.

4 Days to critical

A separate forecast. It does not replace Formula 1.

5 Formula 2

Same line and feeder, next job within 3 km, one possession.

6 CP-SAT

OR-Tools searches a legal window under Rules A, B, and C.

7 Dashboard

Three policies are shown. A person sanctions, shifts, or rejects.

CONTRIBUTION

The combination is the research claim

Shared records

Department exports are normalised into one defect shape. Source severity is kept as reported. It is not rewritten as the urgency score.

Explainable priority

Formula 1 is written down: safety, overdue days, an active speed restriction, and how busy the line is. The screen shows 0–100. The solver can use the unclipped score.

A separate forecast

Gradient boosting estimates days until severity reaches 80. Training labels are synthetic. The forecast is shown beside Formula 1, not inside it.

One possession

Formula 2 packs compatible jobs on the same line and feeder within 3 km. Work time becomes the longest job plus 15 minutes, instead of a fresh visit for each job.

A feasible window

CP-SAT must keep trains and the gang apart, protect Premium and Superfast trains from induced delay, cap other delay, and respect the machine pool and power isolation.

A human decision

Punctuality-First, Asset-First, and Mega-Block reweight the same constraints. AI-1 only prepares evidence-linked rows. It does not choose the block.

PROTOTYPE EVALUATION

What was tested, and what it showed

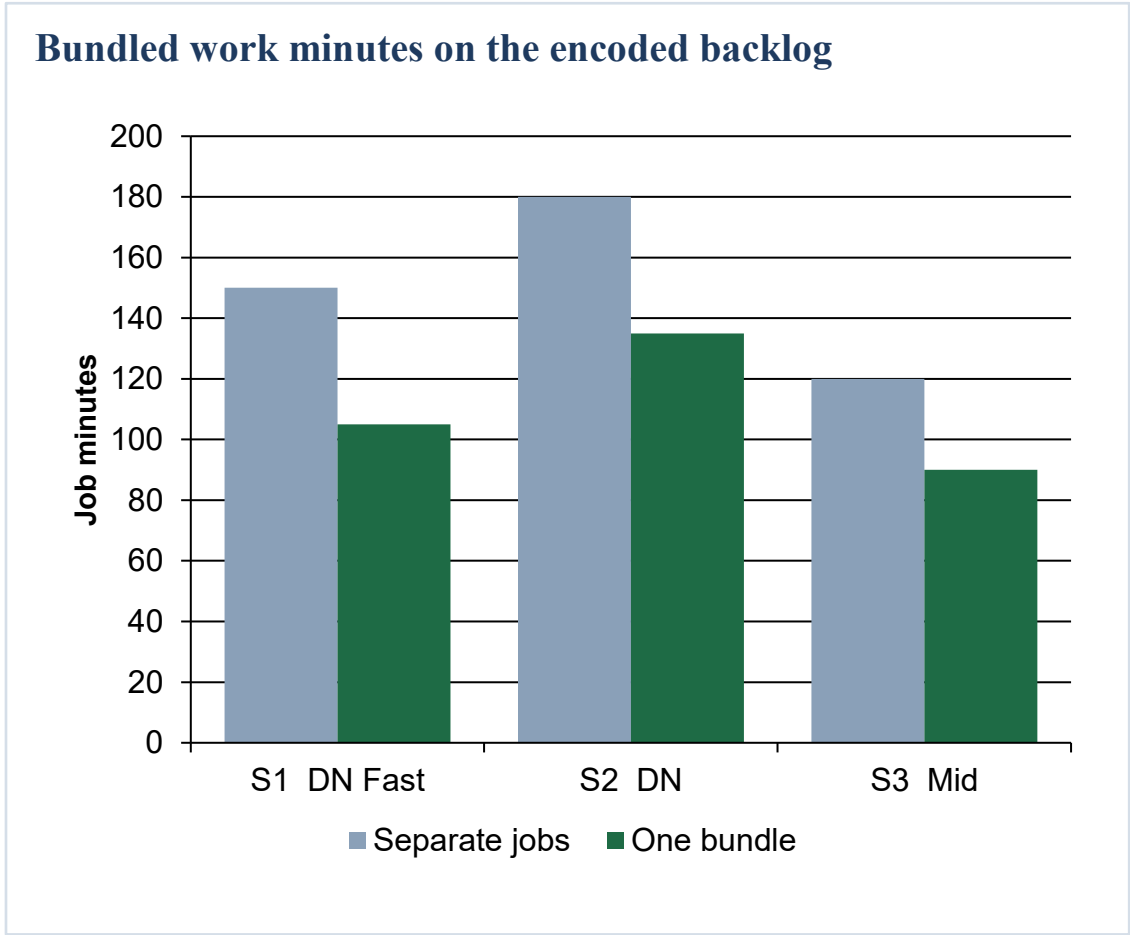
Encoded CSMT–Pune corridor model: 192 km.

24-hour working timetable: 170 modelled services.

Encoded demonstration backlog: 14 defects, 11 bundles.

One 7-day run from 24 September 2026 placed 10 of those 14 defects. Four stayed for a later day. An unsafe window was not forced.

Three multi-defect clusters: 450 job-minutes if done separately, 330 when bundled. Saved: 120 minutes (26.7%).



These are prototype simulations on local demo data. They are not divisional operating results. The live console backlog starts empty until a reviewed report is applied.

CONCLUSION

What this demonstrates, and what it does not

Contribution

The prototype shows a full path from separate maintenance records to a conflict-checked block: validate, review evidence, score urgency, forecast escalation separately, bundle nearby work, then let CP-SAT search a legal window for a person to sanction.

Limits and next evidence

Inputs are local and simulated. AI-1’s ordinary reader is rule-based and misses unfamiliar wording. The forecast is trained on synthetic labels, not real defect histories. A field claim needs authorised feeds and a shadow-mode trial.

[1] Ministry of Railways, SIH 2026 problem statement SIH26027.

[2] Lok Sabha Unstarred Question 3976, answered 12 August 2026. Mail/Express punctuality on Indian Railways.

[3] Rajya Sabha Starred Question 62, answered 5 December 2025. Rolling Block, Gazette notification 30 November 2023.

[4] CAG Report No. 45 of 2017, tabled 13 March 2018. Selected heavy-traffic sections. Historical.